

1. Input

Section, materials and resistance basis exactly as analysed.

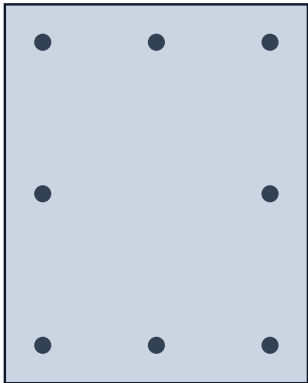
Project information

Project Name	ACI-EB 01 - Rectangle 400 x 500 mm, 8 bars		
Client	Reference benchmark		
Company	P-M Column Designer		
Designed by	Engineering example	Checked by	-
Address	-	Date	2026-08-04

Calculation identity

Section	Benchmark 01 - Rectangle 400 x 50.	Calculation profile	ACI 318-19(22) — Whitney equivalence.
Mechanics	Equivalent rectangular stress block	Resistance profile	ACI CODE-318-19 (Reapproved 202.
Profile edition	2019 (Reapproved 2022)	Verification status	draft
Concrete model basis	Code-default	Method id	aci-318-19-22-global-strength-reduct.
Units	mm · N · MPa; forces reported in kN,.	Sign convention	Compression positive; $M_x = \sum F \cdot (y - y_c)$

Section

Cross-section (mm, about the net concrete centroid)	8 bars	Section properties	
		Gross concrete area A_g	200,000 mm ²
		Reinforcement area A_s	3,200 mm ²
		Reinforcement ratio ρ	1.600 %
		Bar count	8
		Net centroid	(0.0, 0.0) mm
		Bounding box	400.0 × 500.0 mm
		Solids / holes	1 / 0
		Perimeter	1,800.0 mm

Materials

Name	Concrete C40	Standard	ACI318
f _{ck}	40.0 MPa	Model	aci-whitney-block
ε _{cu}	0.00300	α	0.8500
E _c	29,725 MPa	Tension	ignored
Name	Rebar fy 420	Standard	ACI318
f _y	420.0 MPa	Es	200,000 MPa
Model	elastic-perfectly-plastic	ε _y	0.002100
ε _u	not declared	Grades used	1

Design resistance basis

Format	Global resultant strength-reduction f.	Transverse reinforcement	other
φ compression (ties)	0.650	φ compression (spiral)	0.750
φ tension	0.900	ε _{t,limit}	0.00510
Transition rule	yield-plus-strain	Axial cap	0.800 × factored compression pole

Sampling and numerical evidence

Sampling mode	fixed	Active Design points	777
Active Design directions	36	Active Design stations	27
Active Nominal points	902	Active Nominal directions	36
Active Nominal stations	27	Direction tolerance	0.000 %
Worst interpolation error	9.7950 %	Tolerance reached	NO
Concrete discretization	exact polygon clipping	Strain domain	concrete-pivot-ultimate

Scope and limitations

This document reports a preview calculation. It is not an accepted design result and must not be released as a design report.

Resistance is reported in two stages: Nominal is the reference before the resistance treatment, Design is the usable resistance. Only the Design surface is compared with factored ULS demand.

Concrete compression is the code-equivalent rectangular block over $a = \beta_1 \cdot c$, obtained by exact polygon clipping. There is no concrete integration mesh, and the stress between a and c is zero.

Overview interaction diagrams cut the active fixed 27-station, 10° direction surface. An arbitrary exact direction is calculated only when it is requested.

Reinforcement schedule

#	Bar id	Ø (mm)	X (mm)	Y (mm)	As (mm²)	Steel
1	1	22.6	-150.0	-200.0	400.0	Rebar fy 420
2	2	22.6	0.0	-200.0	400.0	Rebar fy 420
3	3	22.6	150.0	-200.0	400.0	Rebar fy 420
4	4	22.6	150.0	0.0	400.0	Rebar fy 420
5	5	22.6	150.0	200.0	400.0	Rebar fy 420
6	6	22.6	0.0	200.0	400.0	Rebar fy 420
7	7	22.6	-150.0	200.0	400.0	Rebar fy 420
8	8	22.6	-150.0	0.0	400.0	Rebar fy 420

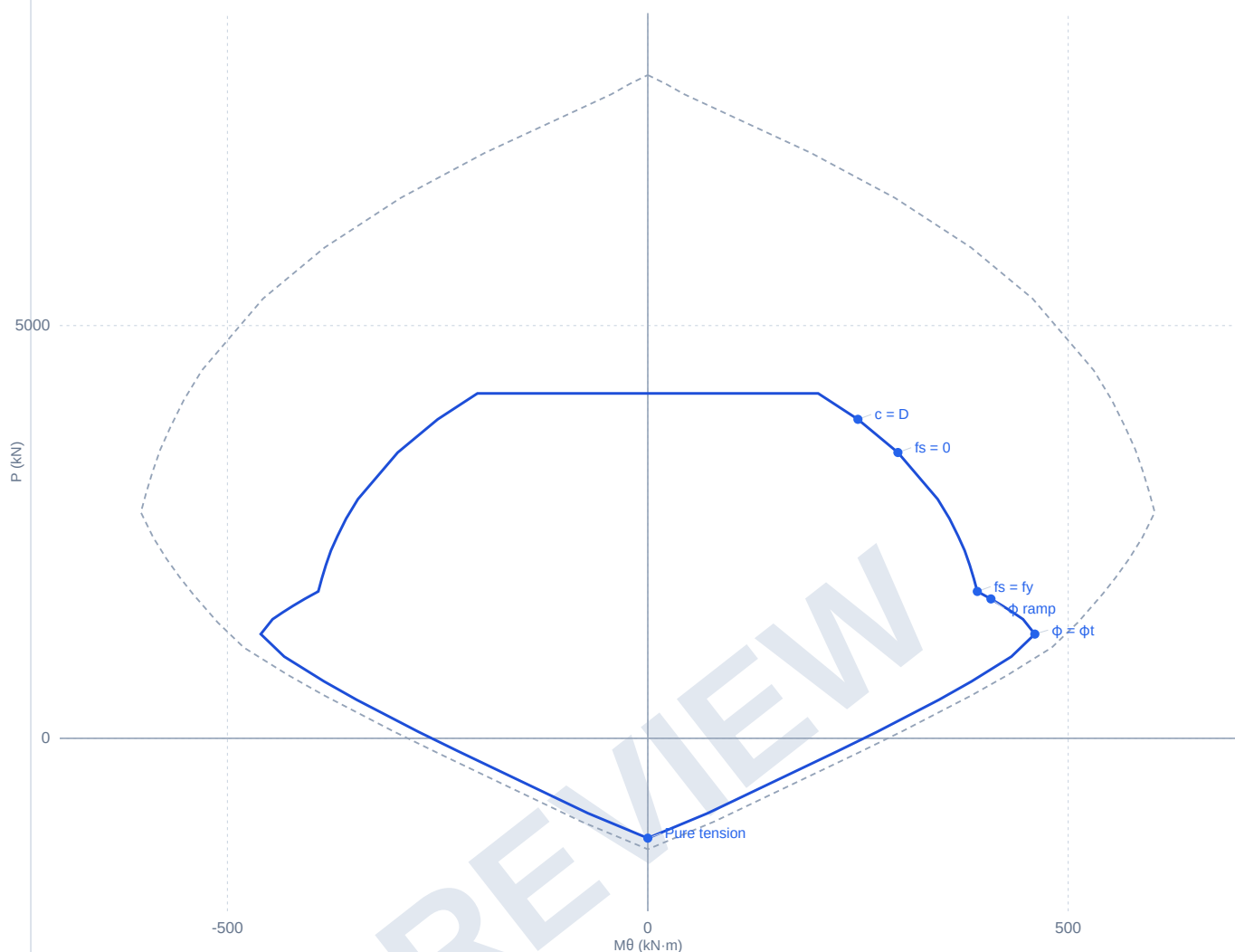
2. Section capacity

Nominal reference and Design resistance. Only Design is compared with factored demand.

Maximum design axial resistance applied at 4178.3 kN; it appears as the flat top of the diagram.

Fixed-grid plane P-M θ at $\theta = 0.00^\circ$

Design solid · Nominal dashed · marked points listed below



Marked states on the diagram

Mark	Station criterion	P design (kN)	M θ design (kN·m)	P nominal (kN)	M θ nominal (kN·m)	ϕ	Class
c = D	c/D = 1	3864.3	249.93	5945.1	384.50	0.650	compression-controlled
fs = 0	$\epsilon_s/\epsilon_y = 0$	3462.4	297.47	5326.8	457.65	0.650	compression-controlled
fs = fy	$\epsilon_s/\epsilon_y = 1$	1779.2	391.93	2737.3	602.98	0.650	compression-controlled
ϕ ramp	$\epsilon_s/\epsilon_y = 1.25$	1688.5	408.14	2433.8	588.30	0.694	transition
$\phi = \phi_t$	$\epsilon_s/\epsilon_y = 2.5$	1262.5	460.35	1402.7	511.50	0.900	tension-controlled
Pure tension	Pure tension	-1209.6	-0.00	-1344.0	-0.00	0.900	compression-controlled

P0 is the compression pole, Pmax the axial cap where one applies, fs = 0 the state where the extreme bar first carries no stress, and fs = fy the balanced state where it first yields. A ϕ mark is the first station of a new resistance class, named in full in the Class column. Every other station on this meridian is tabulated per combination in section 5.

Other published directions (0.00°, 51.25°, 90.00°) are cut from the same surface; each combination is checked in section 4 against the plane it actually lies in.

3. Factored ULS demand

Governing check is the proportional 3D ray against the Design surface.

#	Combination	Pu (kN)	Mux (kN·m)	Muy (kN·m)	θL (°)	φ	Class	UR	Verdict
1	Audit ray 55% - NA 26.93 d.	661.2	97.5	121.5	51.3	0.694	transition	0.550	ADEQUATE
2	Cardinal audit 85% - NA 0 d.	2178.8	0.0	242.0	90.0	0.650	compression-co.	0.850	ADEQUATE
3	Cardinal audit 85% - NA 90 .	1606.2	330.7	0.0	0.0	0.650	compression-co.	0.850	ADEQUATE

UR is the governing proportional utilization: the factored demand vector scaled until it meets the Design surface. A fixed-axial ratio is reported per combination in its detail section and is a secondary diagnostic only.

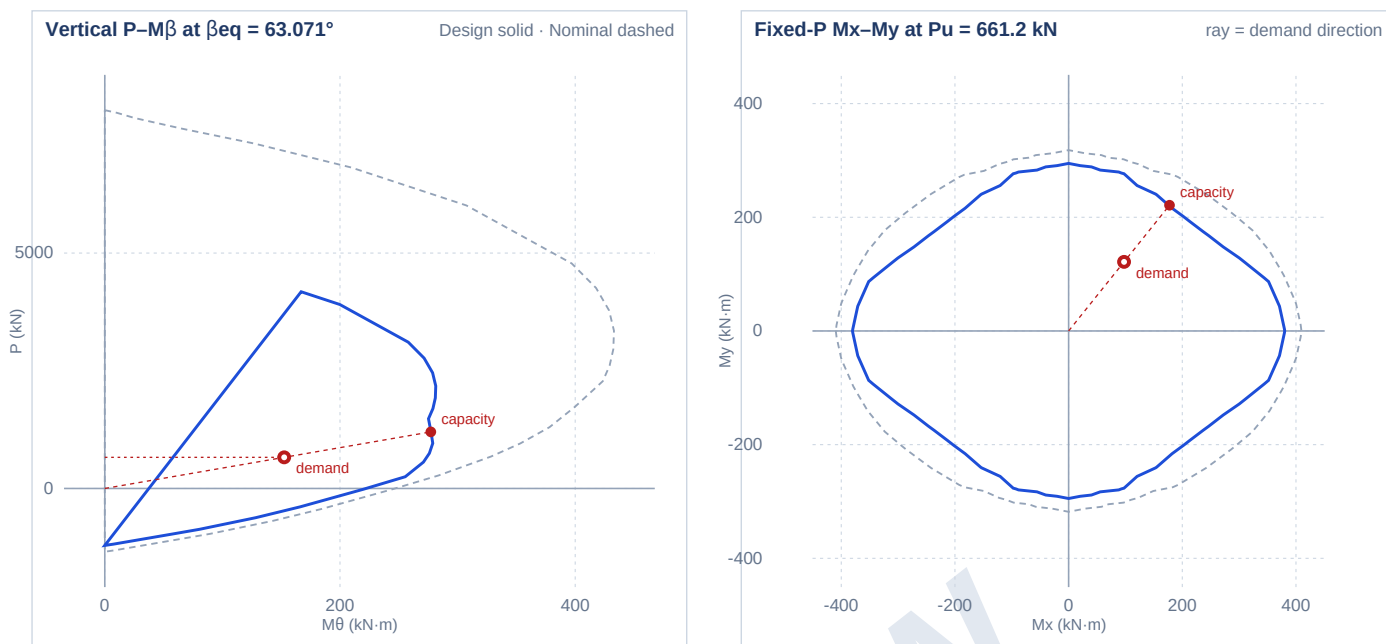
4.1 Audit ray 55% - NA 26.93 deg, c/D 0.55

$P_u = 661.2 \text{ kN}$ · $M_{ux} = 97.5 \text{ kN}\cdot\text{m}$ · $M_{uy} = 121.5 \text{ kN}\cdot\text{m}$ · $\theta_L = 51.25^\circ$

ADEQUATE · utilization 0.5500 · fixed-P diagnostic 0.5491

Factored ULS demand checked by the equivalent-block proportional-ray solver.

Resistance curves through the load point



Left: the exact meridian in the equilibrium strain direction, with the load point projected on it. Right: the surface cut at this combination's own axial force; the dotted ray is the demand moment direction and its intersection is the moment still available there.

Check parameters and solved state

P_u	661.2 kN	M_{ux}	97.5 kN·m
M_{uy}	121.5 kN·m	Moment direction θ_L	51.25°
Resultant moment	155.8 kN·m	Strain-plane ϵ_0	-1.8038e-4
Curvature κ_x	4.9406e-6	Curvature κ_y	9.7262e-6
Strain direction β_{eq}	63.071°	Neutral-axis depth c	275.0 mm
Capacity at P_u	$M_x 177.5 \cdot M_y 221.2 \text{ kN}\cdot\text{m}$	ϕ	0.6940
Classification	transition	Utilization UR	0.5500
Fixed-P ratio	0.5491	Verdict	ADEQUATE

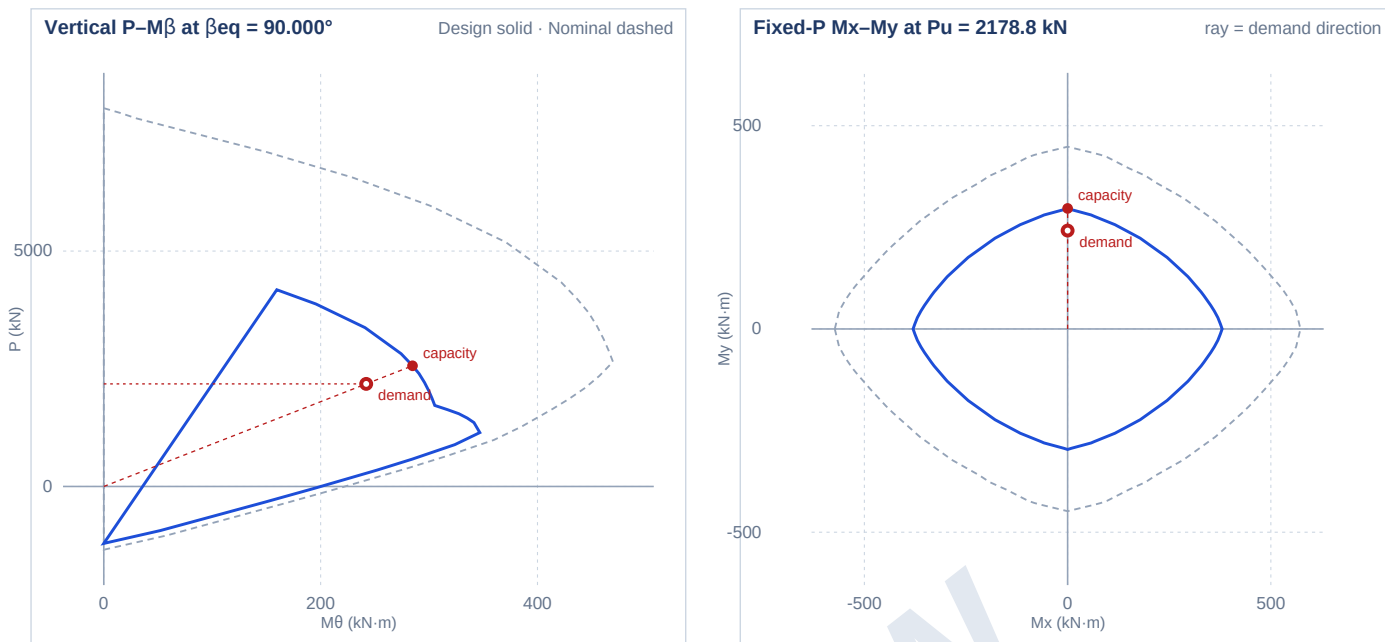
4.2 Cardinal audit 85% - NA 0 deg, c/D 0.55

$P_u = 2178.8 \text{ kN}$ · $M_{ux} = 0.0 \text{ kN}\cdot\text{m}$ · $M_{uy} = 242.0 \text{ kN}\cdot\text{m}$ · $\theta_L = 90.00^\circ$

ADEQUATE · utilization **0.8500** · fixed-P diagnostic **0.8163**

Factored ULS demand checked by the equivalent-block proportional-ray solver.

Resistance curves through the load point



Left: the exact meridian in the equilibrium strain direction, with the load point projected on it. Right: the surface cut at this combination's own axial force; the dotted ray is the demand moment direction and its intersection is the moment still available there.

Check parameters and solved state

P_u	2,178.8 kN	M_{ux}	0.0 kN·m
M_{uy}	242.0 kN·m	Moment direction θ_L	90.00°
Resultant moment	242.0 kN·m	Strain-plane ϵ_0	8.1818e-4
Curvature κ_x	0.0000e+0	Curvature κ_y	1.0909e-5
Strain direction β_{eq}	90.000°	Neutral-axis depth c	275.0 mm
Capacity at P_u	Mx 0.0 · My 296.5 kN·m	ϕ	0.6500
Classification	compression-controlled	Utilization UR	0.8500
Fixed-P ratio	0.8163	Verdict	ADEQUATE

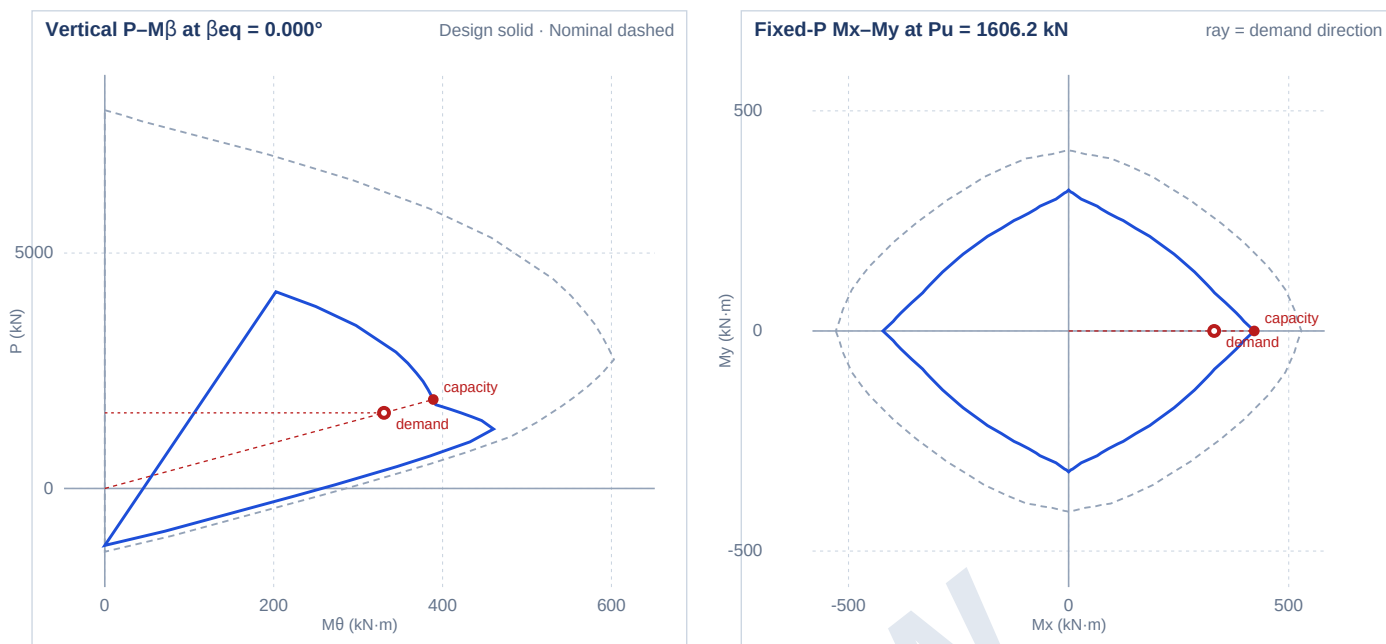
4.3 Cardinal audit 85% - NA 90 deg, c/D 0.55

$P_u = 1606.2 \text{ kN}$ · $M_{ux} = 330.7 \text{ kN}\cdot\text{m}$ · $M_{uy} = 0.0 \text{ kN}\cdot\text{m}$ · $\theta_L = 0.00^\circ$

ADEQUATE · utilization **0.8500** · fixed-P diagnostic **0.7840**

Factored ULS demand checked by the equivalent-block proportional-ray solver.

Resistance curves through the load point



Left: the exact meridian in the equilibrium strain direction, with the load point projected on it. Right: the surface cut at this combination's own axial force; the dotted ray is the demand moment direction and its intersection is the moment still available there.

Check parameters and solved state

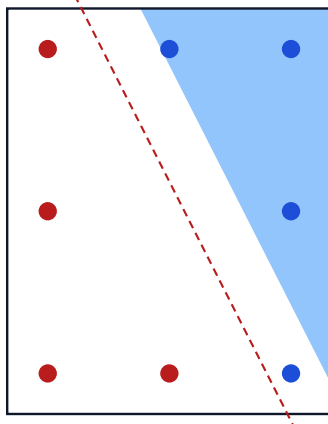
P_u	1,606.2 kN	M_{ux}	330.7 kN·m
M_{uy}	0.0 kN·m	Moment direction θ_L	0.00°
Resultant moment	330.7 kN·m	Strain-plane ϵ_0	2.7273e-4
Curvature κ_x	1.0909e-5	Curvature κ_y	3.0903e-21
Strain direction β_{eq}	0.000°	Neutral-axis depth c	275.0 mm
Capacity at P_u	$M_x 421.7 \cdot M_y 0.0 \text{ kN}\cdot\text{m}$	ϕ	0.6500
Classification	compression-controlled	Utilization UR	0.8500
Fixed-P ratio	0.7840	Verdict	ADEQUATE

5.1 Audit ray 55% - NA 26.93 deg, c/D 0.55

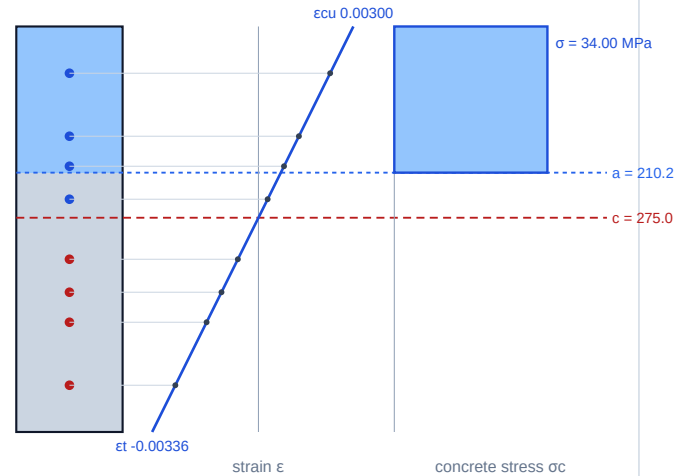
Worked section state and the points behind both curves. $P_u = 661.2 \text{ kN}$ · $M_{ux} = 97.5 \text{ kN}\cdot\text{m}$ · $M_{uy} = 121.5 \text{ kN}\cdot\text{m}$ · $\theta_L = 51.25^\circ$

Section state

Transverse section — neutral axis and compression zone $a=210$



Longitudinal section — strain and stress over the depth $D\theta=583$



Bar fill: blue = compression, red = tension, grey = unstressed. The elevation links each bar to its own strain.

Concrete and resultants

Block area $A_{c,blk}$	54,702.0 mm ²	Block centroid	(121.4, 95.3) mm
Block stress σ_{block}	34.000 MPa	Concrete force C_c	1,859.9 kN
Concrete M_{cx}	177.3 kN·m	Concrete M_{cy}	225.8 kN·m
Nominal P_n	1,732.5 kN	Nominal M_{nx}	255.6 kN·m
Nominal M_{ny}	318.5 kN·m	ϕ	0.6940
Design ϕP_n	1,202.3 kN	Design ϕM_{nx}	177.4 kN·m
Design ϕM_{ny}	221.0 kN·m		

Reinforcement ledger at the solved state

Bar	X (mm)	Y (mm)	A_s (mm ²)	depth (mm)	ϵ	σ_s (MPa)	F (kN)	in block
1	-150.0	-200.0	400.0	515.8	-2.627e-3	-420.00	-168.00	no
2	0.0	-200.0	400.0	382.1	-1.168e-3	-233.70	-93.48	no
3	150.0	-200.0	400.0	248.4	2.904e-4	58.09	23.23	no
4	150.0	0.0	400.0	157.8	1.279e-3	255.71	88.68	yes
5	150.0	200.0	400.0	67.2	2.267e-3	420.00	154.40	yes
6	0.0	200.0	400.0	201.0	8.077e-4	161.55	51.02	yes
7	-150.0	200.0	400.0	334.7	-6.512e-4	-130.24	-52.10	no
8	-150.0	0.0	400.0	425.3	-1.639e-3	-327.86	-131.15	no

Solver and admissibility evidence

Neutral-axis angle θ	26.9290°	Neutral-axis depth c	275.000 mm
Block depth a	210.179 mm	β_1	0.7643
Projected depth $D\theta$	583.07 mm	Controlling ϵ_t	0.002627
Controlling bar	1	Classification	transition
Axial cap governs	no	Component residual	-5.82e-11
Strain admissibility	admissible	Governing utilization	0.5500

Vertical curve points — direct meridian at $\beta_{eq} = 63.071^\circ$

One row per strain-domain station on the meridian the load point is checked against, with the Design and Nominal resultants of that station side by side. P and $M\theta$ are projected on β_{eq} .

#	Criterion	P design (kN)	$M\theta$ design (kN·m)	P nominal (kN)	$M\theta$ nominal (kN·m)
1	axial-cap	4,178.3	166.96	—	—
2	$\epsilon_s/\epsilon_y = 0$	3,907.5	199.86	6,011.6	307.48
3	$\epsilon_s/\epsilon_y = 0.25$	3,107.5	258.04	4,780.7	396.98
4	$\epsilon_s/\epsilon_y = 0.375$	2,768.2	271.53	4,258.7	417.73

#	Criterion	P design (kN)	M ₀ design (kN-m)	P nominal (kN)	M ₀ nominal (kN-m)
5	$\epsilon_s/\epsilon_y = 0.5$	2,457.4	278.73	3,780.6	428.82
6	$\epsilon_s/\epsilon_y = 0.625$	2,177.1	281.43	3,349.4	432.97
7	$\epsilon_s/\epsilon_y = 0.75$	1,922.1	281.07	2,957.1	432.41
8	$\epsilon_s/\epsilon_y = 0.875$	1,697.2	279.03	2,611.0	429.28
9	$\epsilon_s/\epsilon_y = 1$	1,481.3	275.25	2,278.9	423.47
10	$\epsilon_s/\epsilon_y = 1.25$	1,203.5	277.35	1,734.7	399.78
11	$\epsilon_s/\epsilon_y = 1.5$	960.0	278.88	1,301.7	378.15
12	$\epsilon_s/\epsilon_y = 1.75$	747.0	276.00	956.2	353.29
13	$\epsilon_s/\epsilon_y = 2$	559.5	270.93	678.2	328.41
14	$\epsilon_s/\epsilon_y = 2.5$	249.3	255.49	276.9	283.87
15	$\epsilon_s/\epsilon_y = 3$	-15.5	220.04	-17.2	244.49
16	$\epsilon_s/\epsilon_y = 4$	-390.2	166.15	-433.5	184.62
17	$\epsilon_s/\epsilon_y = 5$	-620.6	128.36	-689.5	142.62
18	$\epsilon_s/\epsilon_y = 7.5$	-867.5	80.52	-963.9	89.47
19	$\epsilon_s/\epsilon_y = 10$	-975.6	55.66	-1,084.0	61.85
20	$\epsilon_s/\epsilon_y = 20$	-1,183.4	7.17	-1,314.9	7.97
21	Pure tension	-1,209.6	-0.00	-1,344.0	-0.00
22	Pure compression	—	—	8,035.2	0.00
23	c/D = 3	—	—	8,035.2	0.00
24	c/D = 2	—	—	7,978.9	10.34
25	c/D = 1.5	—	—	7,868.3	25.41
26	c/D = 1.2	—	—	7,643.0	65.53
27	c/D = 1.1	—	—	7,329.1	126.67
28	c/D = 1	—	—	6,818.1	209.48

Fixed-P curve points — Mx-My contour at Pu = 661.2 kN

Cut of the fixed triangulated surface at this combination's own axial force, Design and Nominal paired on the moment direction. The last row repeats the first because the contour is drawn closed.

#	θ M (°)	Mx design	My design	M design	Mx nominal	My nominal	M nominal
1	-180.000	-380.16	-0.00	380.16	-409.61	-0.00	409.61
2	-179.582	-379.82	-2.77	379.83	—	—	—
3	-173.323	-371.01	-43.43	373.54	—	—	—
4	-173.314	-370.98	-43.49	373.52	—	—	—
5	-172.996	-370.11	-45.47	372.90	—	—	—
6	-166.119	-351.74	-86.93	362.32	—	—	—
7	-165.921	-350.52	-87.91	361.37	—	—	—
8	-164.820	-344.10	-93.36	356.54	—	—	—
9	-158.875	-310.80	-120.08	333.19	—	—	—
10	-156.827	-300.19	-128.49	326.53	—	—	—
11	-151.475	-272.03	-147.86	309.62	—	—	—
12	-145.793	-246.54	-167.59	298.11	—	—	—
13	-143.598	-236.86	-174.64	294.28	—	—	—
14	-137.884	-213.10	-192.66	287.28	—	—	—
15	-135.055	-201.83	-201.44	285.15	—	—	—
16	-130.139	-182.06	-215.90	282.42	—	—	—
17	-122.843	-154.74	-239.71	285.31	—	—	—
18	-122.602	-153.72	-240.36	285.31	—	—	—
19	-122.497	-153.33	-240.70	285.39	—	—	—
20	-115.210	-120.45	-255.86	282.80	—	—	—
21	-109.424	-97.52	-276.56	293.25	—	—	—
22	-108.477	-92.84	-277.83	292.93	—	—	—
23	-107.503	-88.18	-279.62	293.20	—	—	—
24	-101.115	-55.61	-283.04	288.46	—	—	—
25	-97.941	-40.26	-288.64	291.43	—	—	—
26	-94.000	-20.34	-290.86	291.57	—	—	—
27	-90.000	0.00	-294.76	294.76	0.00	-318.14	318.14

#	θM (°)	Mx design	My design	M design	Mx nominal	My nominal	M nominal
28	-86.000	20.34	-290.86	291.57	—	—	—
29	-82.059	40.26	-288.64	291.43	—	—	—
30	-78.885	55.61	-283.04	288.46	—	—	—
31	-72.497	88.18	-279.62	293.20	—	—	—
32	-71.523	92.84	-277.83	292.93	—	—	—
33	-70.576	97.52	-276.56	293.25	—	—	—
34	-64.790	120.45	-255.86	282.80	—	—	—
35	-57.503	153.33	-240.70	285.39	—	—	—
36	-57.398	153.72	-240.36	285.31	—	—	—
37	-57.157	154.74	-239.71	285.31	—	—	—
38	-49.861	182.06	-215.90	282.42	—	—	—
39	-44.945	201.83	-201.44	285.15	—	—	—
40	-42.116	213.10	-192.66	287.28	—	—	—
41	-36.402	236.86	-174.64	294.28	—	—	—
42	-34.207	246.54	-167.59	298.11	—	—	—
43	-28.525	272.03	-147.86	309.62	—	—	—
44	-23.173	300.19	-128.49	326.53	—	—	—
45	-21.125	310.80	-120.08	333.19	—	—	—
46	-15.180	344.10	-93.36	356.54	—	—	—
47	-14.079	350.52	-87.91	361.37	—	—	—
48	-13.881	351.74	-86.93	362.32	—	—	—
49	-7.004	370.11	-45.47	372.90	—	—	—
50	-6.686	370.98	-43.49	373.52	—	—	—
51	-6.677	371.01	-43.43	373.54	—	—	—
52	-0.418	379.82	-2.77	379.83	—	—	—
53	0.000	380.16	0.00	380.16	409.61	0.00	409.61
54	0.418	379.82	2.77	379.83	—	—	—
55	6.677	371.01	43.43	373.54	—	—	—
56	6.686	370.98	43.49	373.52	—	—	—
57	7.004	370.11	45.47	372.90	—	—	—
58	13.881	351.74	86.93	362.32	—	—	—
59	14.079	350.52	87.91	361.37	—	—	—
60	15.180	344.10	93.36	356.54	—	—	—
61	21.125	310.80	120.08	333.19	—	—	—
62	23.173	300.19	128.49	326.53	—	—	—
63	28.525	272.03	147.86	309.62	—	—	—
64	34.207	246.54	167.59	298.11	—	—	—
65	36.402	236.86	174.64	294.28	—	—	—
66	42.116	213.10	192.66	287.28	—	—	—
67	44.945	201.83	201.44	285.15	—	—	—
68	49.861	182.06	215.90	282.42	—	—	—
69	57.157	154.74	239.71	285.31	—	—	—
70	57.398	153.72	240.36	285.31	—	—	—
71	57.503	153.33	240.70	285.39	—	—	—
72	64.790	120.45	255.86	282.80	—	—	—
73	70.576	97.52	276.56	293.25	—	—	—
74	71.523	92.84	277.83	292.93	—	—	—
75	72.497	88.18	279.62	293.20	—	—	—
76	78.885	55.61	283.04	288.46	—	—	—
77	82.059	40.26	288.64	291.43	—	—	—
78	86.000	20.34	290.86	291.57	—	—	—
79	90.000	0.00	294.76	294.76	0.00	318.14	318.14
80	94.000	-20.34	290.86	291.57	—	—	—
81	97.941	-40.26	288.64	291.43	—	—	—
82	101.115	-55.61	283.04	288.46	—	—	—
83	107.503	-88.18	279.62	293.20	—	—	—

#	θM (°)	Mx design	My design	M design	Mx nominal	My nominal	M nominal
84	108.477	-92.84	277.83	292.93	—	—	—
85	109.424	-97.52	276.56	293.25	—	—	—
86	115.210	-120.45	255.86	282.80	—	—	—
87	122.497	-153.33	240.70	285.39	—	—	—
88	122.602	-153.72	240.36	285.31	—	—	—
89	122.843	-154.74	239.71	285.31	—	—	—
90	130.139	-182.06	215.90	282.42	—	—	—
91	135.055	-201.83	201.44	285.15	—	—	—
92	137.884	-213.10	192.66	287.28	—	—	—
93	143.598	-236.86	174.64	294.28	—	—	—
94	145.793	-246.54	167.59	298.11	—	—	—
95	151.475	-272.03	147.86	309.62	—	—	—
96	156.827	-300.19	128.49	326.53	—	—	—
97	158.875	-310.80	120.08	333.19	—	—	—
98	164.820	-344.10	93.36	356.54	—	—	—
99	165.921	-350.52	87.91	361.37	—	—	—
100	166.119	-351.74	86.93	362.32	—	—	—
101	172.996	-370.11	45.47	372.90	—	—	—
102	173.314	-370.98	43.49	373.52	—	—	—
103	173.323	-371.01	43.43	373.54	—	—	—
104	179.582	-379.82	2.77	379.83	—	—	—
105	-180.000	-380.16	-0.00	380.16	-409.61	-0.00	409.61
106	-178.556	—	—	—	-408.34	-10.29	408.47
107	-173.229	—	—	—	-400.27	-47.52	403.09
108	-171.664	—	—	—	-396.13	-58.04	400.36
109	-165.610	—	—	—	-378.93	-97.22	391.20
110	-164.107	—	—	—	-373.76	-106.42	388.62
111	-158.304	—	—	—	-353.43	-140.61	380.37
112	-151.999	—	—	—	-326.39	-173.55	369.66
113	-151.319	—	—	—	-323.59	-177.02	368.85
114	-145.831	—	—	—	-295.54	-200.61	357.19
115	-143.880	—	—	—	-285.83	-208.58	353.85
116	-143.559	—	—	—	-284.12	-209.79	353.18
117	-138.742	—	—	—	-259.78	-227.89	345.57
118	-135.217	—	—	—	-242.47	-240.64	341.62
119	-135.089	—	—	—	-241.85	-241.11	341.50
120	-125.429	—	—	—	-192.61	-270.74	332.26
121	-123.682	—	—	—	-183.23	-274.92	330.39
122	-117.893	—	—	—	-149.03	-281.56	318.57
123	-112.608	—	—	—	-122.28	-293.64	318.08
124	-110.229	—	—	—	-109.49	-297.12	316.65
125	-107.763	—	—	—	-96.75	-302.01	317.13
126	-102.978	—	—	—	-70.24	-304.80	312.79
127	-100.574	—	—	—	-57.75	-309.35	314.70
128	-95.333	—	—	—	-29.17	-312.53	313.89
129	-84.667	—	—	—	29.17	-312.53	313.89
130	-79.426	—	—	—	57.75	-309.35	314.70
131	-77.022	—	—	—	70.24	-304.80	312.79
132	-72.237	—	—	—	96.75	-302.01	317.13
133	-69.771	—	—	—	109.49	-297.12	316.65
134	-67.392	—	—	—	122.28	-293.64	318.08
135	-62.107	—	—	—	149.03	-281.56	318.57
136	-56.318	—	—	—	183.23	-274.92	330.39
137	-54.571	—	—	—	192.61	-270.74	332.26
138	-44.911	—	—	—	241.85	-241.11	341.50
139	-44.783	—	—	—	242.47	-240.64	341.62

#	θ_M (°)	Mx design	My design	M design	Mx nominal	My nominal	M nominal
140	-41.258	—	—	—	259.78	-227.89	345.57
141	-36.441	—	—	—	284.12	-209.79	353.18
142	-36.120	—	—	—	285.83	-208.58	353.85
143	-34.169	—	—	—	295.54	-200.61	357.19
144	-28.681	—	—	—	323.59	-177.02	368.85
145	-28.001	—	—	—	326.39	-173.55	369.66
146	-21.696	—	—	—	353.43	-140.61	380.37
147	-15.893	—	—	—	373.76	-106.42	388.62
148	-14.390	—	—	—	378.93	-97.22	391.20
149	-8.336	—	—	—	396.13	-58.04	400.36
150	-6.771	—	—	—	400.27	-47.52	403.09
151	-1.444	—	—	—	408.34	-10.29	408.47
152	1.444	—	—	—	408.34	10.29	408.47
153	6.771	—	—	—	400.27	47.52	403.09
154	8.336	—	—	—	396.13	58.04	400.36
155	14.390	—	—	—	378.93	97.22	391.20
156	15.893	—	—	—	373.76	106.42	388.62
157	21.696	—	—	—	353.43	140.61	380.37
158	28.001	—	—	—	326.39	173.55	369.66
159	28.681	—	—	—	323.59	177.02	368.85
160	34.169	—	—	—	295.54	200.61	357.19
161	36.120	—	—	—	285.83	208.58	353.85
162	36.441	—	—	—	284.12	209.79	353.18
163	41.258	—	—	—	259.78	227.89	345.57
164	44.783	—	—	—	242.47	240.64	341.62
165	44.911	—	—	—	241.85	241.11	341.50
166	54.571	—	—	—	192.61	270.74	332.26
167	56.318	—	—	—	183.23	274.92	330.39
168	62.107	—	—	—	149.03	281.56	318.57
169	67.392	—	—	—	122.28	293.64	318.08
170	69.771	—	—	—	109.49	297.12	316.65
171	72.237	—	—	—	96.75	302.01	317.13
172	77.022	—	—	—	70.24	304.80	312.79
173	79.426	—	—	—	57.75	309.35	314.70
174	84.667	—	—	—	29.17	312.53	313.89
175	95.333	—	—	—	-29.17	312.53	313.89
176	100.574	—	—	—	-57.75	309.35	314.70
177	102.978	—	—	—	-70.24	304.80	312.79
178	107.763	—	—	—	-96.75	302.01	317.13
179	110.229	—	—	—	-109.49	297.12	316.65
180	112.608	—	—	—	-122.28	293.64	318.08
181	117.893	—	—	—	-149.03	281.56	318.57
182	123.682	—	—	—	-183.23	274.92	330.39
183	125.429	—	—	—	-192.61	270.74	332.26
184	135.089	—	—	—	-241.85	241.11	341.50
185	135.217	—	—	—	-242.47	240.64	341.62
186	138.742	—	—	—	-259.78	227.89	345.57
187	143.559	—	—	—	-284.12	209.79	353.18
188	143.880	—	—	—	-285.83	208.58	353.85
189	145.831	—	—	—	-295.54	200.61	357.19
190	151.319	—	—	—	-323.59	177.02	368.85
191	151.999	—	—	—	-326.39	173.55	369.66
192	158.304	—	—	—	-353.43	140.61	380.37
193	164.107	—	—	—	-373.76	106.42	388.62
194	165.610	—	—	—	-378.93	97.22	391.20
195	171.664	—	—	—	-396.13	58.04	400.36

P-M Column Designer | ACI-EB 01 - Rectangle 400 x 500 mm, 8 bars | Reference benchmark | Benchm. not stamped (deterministic output)

#	θ_M (°)	Mx design	My design	M design	Mx nominal	My nominal	M nominal
196	173.229	—	—	—	-400.27	47.52	403.09
197	178.556	—	—	—	-408.34	10.29	408.47

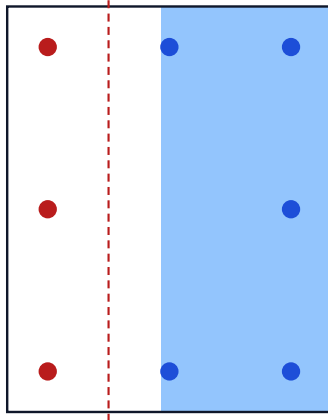
PREVIEW

5.2 Cardinal audit 85% - NA 0 deg, c/D 0.55

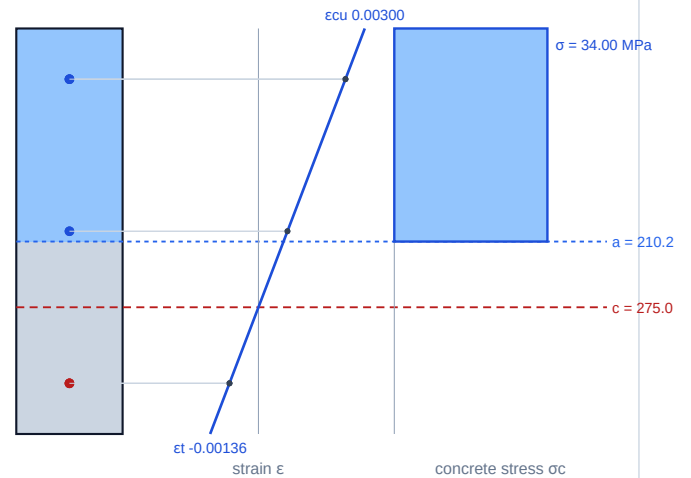
Worked section state and the points behind both curves. $P_u = 2178.8 \text{ kN}$ · $M_{ux} = 0.0 \text{ kN}\cdot\text{m}$ · $M_{uy} = 242.0 \text{ kN}\cdot\text{m}$ · $\theta_L = 90.00^\circ$

Section state

Transverse section — neutral axis and compression zone $a=210$



Longitudinal section — strain and stress over the depth $D\theta=400$



Bar fill: blue = compression, red = tension, grey = unstressed. The elevation links each bar to its own strain.

Concrete and resultants

Block area $A_{c,blk}$	105,089.3 mm ²	Block centroid	(94.9, -0.0) mm
Block stress σ_{block}	34.000 MPa	Concrete force C_c	3,573.0 kN
Concrete M_{cx}	-0.0 kN·m	Concrete M_{cy}	339.1 kN·m
Nominal P_n	3,943.6 kN	Nominal M_{nx}	-0.0 kN·m
Nominal M_{ny}	438.1 kN·m	ϕ	0.6500
Design ϕP_n	2,563.3 kN	Design ϕM_{nx}	-0.0 kN·m
Design ϕM_{ny}	284.7 kN·m		

Reinforcement ledger at the solved state

Bar	X (mm)	Y (mm)	A_s (mm ²)	depth (mm)	ϵ	σ_s (MPa)	F (kN)	in block
1	-150.0	-200.0	400.0	350.0	-8.182e-4	-163.64	-65.45	no
2	0.0	-200.0	400.0	200.0	8.182e-4	163.64	51.85	yes
3	150.0	-200.0	400.0	50.0	2.455e-3	420.00	154.40	yes
4	150.0	0.0	400.0	50.0	2.455e-3	420.00	154.40	yes
5	150.0	200.0	400.0	50.0	2.455e-3	420.00	154.40	yes
6	0.0	200.0	400.0	200.0	8.182e-4	163.64	51.85	yes
7	-150.0	200.0	400.0	350.0	-8.182e-4	-163.64	-65.45	no
8	-150.0	0.0	400.0	350.0	-8.182e-4	-163.64	-65.45	no

Solver and admissibility evidence

Neutral-axis angle θ	0.0000°	Neutral-axis depth c	275.000 mm
Block depth a	210.179 mm	β_1	0.7643
Projected depth $D\theta$	400.00 mm	Controlling ϵ_t	0.000818
Controlling bar	1	Classification	compression-controlled
Axial cap governs	no	Component residual	-1.75e-10
Strain admissibility	admissible	Governing utilization	0.8500

Vertical curve points — direct meridian at $\beta_{eq} = 90.000^\circ$

One row per strain-domain station on the meridian the load point is checked against, with the Design and Nominal resultants of that station side by side. P and $M\theta$ are projected on β_{eq} .

#	Criterion	P design (kN)	$M\theta$ design (kN·m)	P nominal (kN)	$M\theta$ nominal (kN·m)
1	axial-cap	4,178.3	159.56	—	—
2	$c/D = 1$	3,876.0	195.64	5,963.1	300.99
3	$\epsilon_s/\epsilon_y = 0$	3,373.0	240.99	5,189.2	370.75
4	$\epsilon_s/\epsilon_y = 0.25$	2,819.7	274.22	4,337.9	421.88

#	Criterion	P design (kN)	M ₀ design (kN-m)	P nominal (kN)	M ₀ nominal (kN-m)
5	$\epsilon_s/\epsilon_y = 0.375$	2,588.8	283.81	3,982.7	436.63
6	$\epsilon_s/\epsilon_y = 0.5$	2,398.1	290.71	3,689.4	447.25
7	$\epsilon_s/\epsilon_y = 0.625$	2,208.3	295.81	3,397.4	455.08
8	$\epsilon_s/\epsilon_y = 0.75$	2,033.8	299.68	3,128.9	461.04
9	$\epsilon_s/\epsilon_y = 0.875$	1,872.0	302.73	2,880.1	465.74
10	$\epsilon_s/\epsilon_y = 1$	1,721.1	305.25	2,647.9	469.62
11	$\epsilon_s/\epsilon_y = 1.25$	1,630.5	317.14	2,350.2	457.14
12	$\epsilon_s/\epsilon_y = 1.5$	1,545.2	327.46	2,095.2	444.01
13	$\epsilon_s/\epsilon_y = 1.75$	1,453.0	335.20	1,859.9	429.06
14	$\epsilon_s/\epsilon_y = 2$	1,357.0	341.40	1,644.8	413.81
15	$\epsilon_s/\epsilon_y = 2.5$	1,144.5	346.82	1,271.7	385.35
16	$\epsilon_s/\epsilon_y = 3$	888.6	323.77	987.3	359.75
17	$\epsilon_s/\epsilon_y = 4$	580.5	284.47	645.1	316.07
18	$\epsilon_s/\epsilon_y = 5$	348.2	252.11	386.9	280.13
19	$\epsilon_s/\epsilon_y = 7.5$	-31.7	195.41	-35.3	217.12
20	$\epsilon_s/\epsilon_y = 10$	-337.0	147.92	-374.4	164.36
21	$\epsilon_s/\epsilon_y = 20$	-936.7	52.14	-1,040.8	57.93
22	Pure tension	-1,209.6	-0.00	-1,344.0	-0.00
23	Pure compression	—	—	8,035.2	0.00
24	c/D = 3	—	—	8,035.2	0.00
25	c/D = 2	—	—	7,936.2	14.85
26	c/D = 1.5	—	—	7,815.2	30.60
27	c/D = 1.2	—	—	7,106.8	149.70
28	c/D = 1.1	—	—	6,561.9	229.51

Fixed-P curve points — Mx-My contour at Pu = 2,178.8 kN

Cut of the fixed triangulated surface at this combination's own axial force, Design and Nominal paired on the moment direction. The last row repeats the first because the contour is drawn closed.

#	θM (°)	Mx design	My design	M design	Mx nominal	My nominal	M nominal
1	-180.000	-380.16	-0.00	380.16	-572.74	-0.00	572.74
2	-178.128	-375.82	-12.29	376.02	—	—	—
3	-175.688	-369.64	-27.87	370.69	—	—	—
4	-173.047	-360.91	-44.01	363.59	—	—	—
5	-170.621	-352.00	-58.14	356.77	—	—	—
6	-166.311	-335.38	-81.69	345.19	—	—	—
7	-164.513	-328.41	-91.00	340.79	—	—	—
8	-157.218	-298.83	-125.50	324.11	—	—	—
9	-156.420	-295.64	-129.04	322.57	—	—	—
10	-149.983	-267.84	-154.75	309.33	—	—	—
11	-144.229	-244.51	-176.16	301.36	—	—	—
12	-143.466	-241.06	-178.60	300.02	—	—	—
13	-128.619	-178.43	-223.36	285.88	—	—	—
14	-128.166	-176.39	-224.42	285.45	—	—	—
15	-122.987	-153.63	-236.69	282.18	—	—	—
16	-115.203	-119.86	-254.69	281.49	—	—	—
17	-114.513	-116.90	-256.35	281.75	—	—	—
18	-103.367	-65.81	-276.96	284.68	—	—	—
19	-101.685	-57.99	-280.38	286.31	—	—	—
20	-91.698	-8.71	-293.71	293.84	—	—	—
21	-90.000	0.00	-296.46	296.46	0.00	-448.31	448.31
22	-88.302	8.71	-293.71	293.84	—	—	—
23	-78.315	57.99	-280.38	286.31	—	—	—
24	-76.633	65.81	-276.96	284.68	—	—	—
25	-65.487	116.90	-256.35	281.75	—	—	—
26	-64.797	119.86	-254.69	281.49	—	—	—
27	-57.013	153.63	-236.69	282.18	—	—	—

#	θ_M (°)	Mx design	My design	M design	Mx nominal	My nominal	M nominal
28	-51.834	176.39	-224.42	285.45	—	—	—
29	-51.381	178.43	-223.36	285.88	—	—	—
30	-36.534	241.06	-178.60	300.02	—	—	—
31	-35.771	244.51	-176.16	301.36	—	—	—
32	-30.017	267.84	-154.75	309.33	—	—	—
33	-23.580	295.64	-129.04	322.57	—	—	—
34	-22.782	298.83	-125.50	324.11	—	—	—
35	-15.487	328.41	-91.00	340.79	—	—	—
36	-13.689	335.38	-81.69	345.19	—	—	—
37	-9.379	352.00	-58.14	356.77	—	—	—
38	-6.953	360.91	-44.01	363.59	—	—	—
39	-4.312	369.64	-27.87	370.69	—	—	—
40	-1.872	375.82	-12.29	376.02	—	—	—
41	-0.000	380.16	-0.00	380.16	—	—	—
42	1.872	375.82	12.29	376.02	—	—	—
43	4.312	369.64	27.87	370.69	—	—	—
44	6.953	360.91	44.01	363.59	—	—	—
45	9.379	352.00	58.14	356.77	—	—	—
46	13.689	335.38	81.69	345.19	—	—	—
47	15.487	328.41	91.00	340.79	—	—	—
48	22.782	298.83	125.50	324.11	—	—	—
49	23.580	295.64	129.04	322.57	—	—	—
50	30.017	267.84	154.75	309.33	—	—	—
51	35.771	244.51	176.16	301.36	—	—	—
52	36.534	241.06	178.60	300.02	—	—	—
53	51.381	178.43	223.36	285.88	—	—	—
54	51.834	176.39	224.42	285.45	—	—	—
55	57.013	153.63	236.69	282.18	—	—	—
56	64.797	119.86	254.69	281.49	—	—	—
57	65.487	116.90	256.35	281.75	—	—	—
58	76.633	65.81	276.96	284.68	—	—	—
59	78.315	57.99	280.38	286.31	—	—	—
60	88.302	8.71	293.71	293.84	—	—	—
61	90.000	0.00	296.46	296.46	0.00	448.31	448.31
62	91.698	-8.71	293.71	293.84	—	—	—
63	101.685	-57.99	280.38	286.31	—	—	—
64	103.367	-65.81	276.96	284.68	—	—	—
65	114.513	-116.90	256.35	281.75	—	—	—
66	115.203	-119.86	254.69	281.49	—	—	—
67	122.987	-153.63	236.69	282.18	—	—	—
68	128.166	-176.39	224.42	285.45	—	—	—
69	128.619	-178.43	223.36	285.88	—	—	—
70	143.466	-241.06	178.60	300.02	—	—	—
71	144.229	-244.51	176.16	301.36	—	—	—
72	149.983	-267.84	154.75	309.33	—	—	—
73	156.420	-295.64	129.04	322.57	—	—	—
74	157.218	-298.83	125.50	324.11	—	—	—
75	164.513	-328.41	91.00	340.79	—	—	—
76	166.311	-335.38	81.69	345.19	—	—	—
77	170.621	-352.00	58.14	356.77	—	—	—
78	173.047	-360.91	44.01	363.59	—	—	—
79	175.688	-369.64	27.87	370.69	—	—	—
80	178.128	-375.82	12.29	376.02	—	—	—
81	-180.000	-380.16	-0.00	380.16	-572.74	-0.00	572.74
82	-179.789	—	—	—	-572.29	-2.10	572.29
83	-175.528	—	—	—	-562.11	-43.96	563.82

#	θ_M (°)	Mx design	My design	M design	Mx nominal	My nominal	M nominal
84	-172.568	—	—	—	-546.74	-71.32	551.38
85	-169.789	—	—	—	-530.18	-95.50	538.71
86	-168.304	—	—	—	-520.35	-107.72	531.38
87	-162.934	—	—	—	-484.67	-148.79	506.99
88	-157.042	—	—	—	-447.07	-189.38	485.52
89	-154.343	—	—	—	-429.60	-206.35	476.59
90	-144.045	—	—	—	-364.55	-264.42	450.35
91	-143.841	—	—	—	-363.29	-265.48	449.96
92	-132.583	—	—	—	-291.51	-317.20	430.81
93	-131.629	—	—	—	-285.73	-321.50	430.12
94	-121.158	—	—	—	-217.53	-359.78	420.43
95	-116.997	—	—	—	-191.92	-376.73	422.80
96	-110.961	—	—	—	-151.32	-394.99	422.98
97	-102.909	—	—	—	-96.92	-422.85	433.82
98	-100.809	—	—	—	-81.74	-428.12	435.85
99	-98.507	—	—	—	-64.72	-432.66	437.48
100	-93.412	—	—	—	-26.35	-441.88	442.66
101	-86.588	—	—	—	26.35	-441.88	442.66
102	-81.493	—	—	—	64.72	-432.66	437.48
103	-79.191	—	—	—	81.74	-428.12	435.85
104	-77.091	—	—	—	96.92	-422.85	433.82
105	-69.039	—	—	—	151.32	-394.99	422.98
106	-63.003	—	—	—	191.92	-376.73	422.80
107	-58.842	—	—	—	217.53	-359.78	420.43
108	-48.371	—	—	—	285.73	-321.50	430.12
109	-47.417	—	—	—	291.51	-317.20	430.81
110	-36.159	—	—	—	363.29	-265.48	449.96
111	-35.955	—	—	—	364.55	-264.42	450.35
112	-25.657	—	—	—	429.60	-206.35	476.59
113	-22.958	—	—	—	447.07	-189.38	485.52
114	-17.066	—	—	—	484.67	-148.79	506.99
115	-11.696	—	—	—	520.35	-107.72	531.38
116	-10.211	—	—	—	530.18	-95.50	538.71
117	-7.432	—	—	—	546.74	-71.32	551.38
118	-4.472	—	—	—	562.11	-43.96	563.82
119	-0.211	—	—	—	572.29	-2.10	572.29
120	0.000	—	—	—	572.74	0.00	572.74
121	0.211	—	—	—	572.29	2.10	572.29
122	4.472	—	—	—	562.11	43.96	563.82
123	7.432	—	—	—	546.74	71.32	551.38
124	10.211	—	—	—	530.18	95.50	538.71
125	11.696	—	—	—	520.35	107.72	531.38
126	17.066	—	—	—	484.67	148.79	506.99
127	22.958	—	—	—	447.07	189.38	485.52
128	25.657	—	—	—	429.60	206.35	476.59
129	35.955	—	—	—	364.55	264.42	450.35
130	36.159	—	—	—	363.29	265.48	449.96
131	47.417	—	—	—	291.51	317.20	430.81
132	48.371	—	—	—	285.73	321.50	430.12
133	58.842	—	—	—	217.53	359.78	420.43
134	63.003	—	—	—	191.92	376.73	422.80
135	69.039	—	—	—	151.32	394.99	422.98
136	77.091	—	—	—	96.92	422.85	433.82
137	79.191	—	—	—	81.74	428.12	435.85
138	81.493	—	—	—	64.72	432.66	437.48
139	86.588	—	—	—	26.35	441.88	442.66

#	θ_M (°)	Mx design	My design	M design	Mx nominal	My nominal	M nominal
140	93.412	—	—	—	-26.35	441.88	442.66
141	98.507	—	—	—	-64.72	432.66	437.48
142	100.809	—	—	—	-81.74	428.12	435.85
143	102.909	—	—	—	-96.92	422.85	433.82
144	110.961	—	—	—	-151.32	394.99	422.98
145	116.997	—	—	—	-191.92	376.73	422.80
146	121.158	—	—	—	-217.53	359.78	420.43
147	131.629	—	—	—	-285.73	321.50	430.12
148	132.583	—	—	—	-291.51	317.20	430.81
149	143.841	—	—	—	-363.29	265.48	449.96
150	144.045	—	—	—	-364.55	264.42	450.35
151	154.343	—	—	—	-429.60	206.35	476.59
152	157.042	—	—	—	-447.07	189.38	485.52
153	162.934	—	—	—	-484.67	148.79	506.99
154	168.304	—	—	—	-520.35	107.72	531.38
155	169.789	—	—	—	-530.18	95.50	538.71
156	172.568	—	—	—	-546.74	71.32	551.38
157	175.528	—	—	—	-562.11	43.96	563.82
158	179.789	—	—	—	-572.29	2.10	572.29

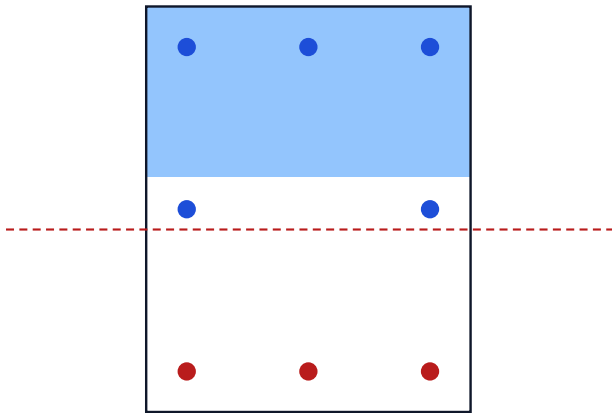
PREVIEW

5.3 Cardinal audit 85% - NA 90 deg, c/D 0.55

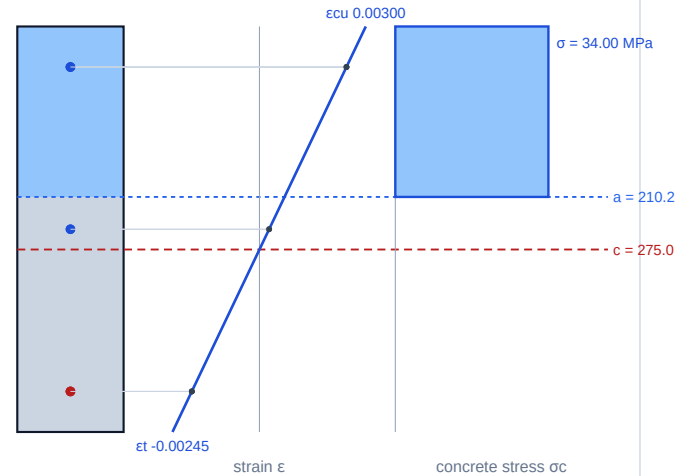
Worked section state and the points behind both curves. $P_u = 1606.2 \text{ kN}$ · $M_{ux} = 330.7 \text{ kN}\cdot\text{m}$ · $M_{uy} = 0.0 \text{ kN}\cdot\text{m}$ · $\theta_L = 0.00^\circ$

Section state

Transverse section — neutral axis and compression zone $a=210$



Longitudinal section — strain and stress over the depth $D\theta=500$



Bar fill: blue = compression, red = tension, grey = unstressed. The elevation links each bar to its own strain.

Concrete and resultants

Block area $A_{c,blk}$	84,071.4 mm²	Block centroid	(0.0, 144.9) mm
Block stress σ_{block}	34.000 MPa	Concrete force C_c	2,858.4 kN
Concrete M_{cx}	414.2 kN·m	Concrete M_{cy}	0.0 kN·m
Nominal P_n	2,907.1 kN	Nominal M_{nx}	598.5 kN·m
Nominal M_{ny}	0.0 kN·m	ϕ	0.6500
Design ϕP_n	1,889.6 kN	Design ϕM_{nx}	389.0 kN·m
Design ϕM_{ny}	0.0 kN·m		

Reinforcement ledger at the solved state

Bar	X (mm)	Y (mm)	A_s (mm ²)	depth (mm)	ϵ	σ_s (MPa)	F (kN)	in block
1	-150.0	-200.0	400.0	450.0	-1.909e-3	-381.82	-152.73	no
2	0.0	-200.0	400.0	450.0	-1.909e-3	-381.82	-152.73	no
3	150.0	-200.0	400.0	450.0	-1.909e-3	-381.82	-152.73	no
4	150.0	0.0	400.0	250.0	2.727e-4	54.55	21.82	no
5	150.0	200.0	400.0	50.0	2.455e-3	420.00	154.40	yes
6	0.0	200.0	400.0	50.0	2.455e-3	420.00	154.40	yes
7	-150.0	200.0	400.0	50.0	2.455e-3	420.00	154.40	yes
8	-150.0	0.0	400.0	250.0	2.727e-4	54.55	21.82	no

Solver and admissibility evidence

Neutral-axis angle θ	90.0000°	Neutral-axis depth c	275.000 mm
Block depth a	210.179 mm	β_1	0.7643
Projected depth $D\theta$	500.00 mm	Controlling ϵ_t	0.001909
Controlling bar	1	Classification	compression-controlled
Axial cap governs	no	Component residual	-7.28e-11
Strain admissibility	admissible	Governing utilization	0.8500

Vertical curve points — direct meridian at $\beta_{eq} = 0.000^\circ$

One row per strain-domain station on the meridian the load point is checked against, with the Design and Nominal resultants of that station side by side. P and $M\theta$ are projected on β_{eq} .

#	Criterion	P design (kN)	$M\theta$ design (kN·m)	P nominal (kN)	$M\theta$ nominal (kN·m)
1	axial-cap	4,178.3	202.66	—	—
2	c/D = 1	3,864.3	249.93	5,945.1	384.50
3	$\epsilon_s/\epsilon_y = 0$	3,462.4	297.47	5,326.8	457.65
4	$\epsilon_s/\epsilon_y = 0.25$	2,897.3	344.79	4,457.5	530.44

#	Criterion	P design (kN)	M ₀ design (kN-m)	P nominal (kN)	M ₀ nominal (kN-m)
5	$\epsilon_s/\epsilon_y = 0.375$	2,661.9	358.81	4,095.2	552.02
6	$\epsilon_s/\epsilon_y = 0.5$	2,449.7	369.13	3,768.8	567.89
7	$\epsilon_s/\epsilon_y = 0.625$	2,274.2	376.91	3,498.7	579.85
8	$\epsilon_s/\epsilon_y = 0.75$	2,096.7	382.96	3,225.7	589.17
9	$\epsilon_s/\epsilon_y = 0.875$	1,932.4	387.84	2,972.9	596.68
10	$\epsilon_s/\epsilon_y = 1$	1,779.2	391.93	2,737.3	602.98
11	$\epsilon_s/\epsilon_y = 1.25$	1,688.5	408.14	2,433.8	588.30
12	$\epsilon_s/\epsilon_y = 1.5$	1,603.5	422.19	2,174.2	572.46
13	$\epsilon_s/\epsilon_y = 1.75$	1,521.9	434.78	1,948.1	556.52
14	$\epsilon_s/\epsilon_y = 2$	1,442.2	446.35	1,748.1	541.03
15	$\epsilon_s/\epsilon_y = 2.5$	1,262.5	460.35	1,402.7	511.50
16	$\epsilon_s/\epsilon_y = 3$	990.0	432.50	1,100.0	480.55
17	$\epsilon_s/\epsilon_y = 4$	689.5	385.08	766.1	427.86
18	$\epsilon_s/\epsilon_y = 5$	466.8	346.30	518.6	384.78
19	$\epsilon_s/\epsilon_y = 7.5$	78.8	272.83	87.6	303.15
20	$\epsilon_s/\epsilon_y = 10$	-157.8	225.36	-175.3	250.40
21	$\epsilon_s/\epsilon_y = 20$	-907.4	71.26	-1,008.2	79.18
22	Pure tension	-1,209.6	-0.00	-1,344.0	-0.00
23	Pure compression	—	—	8,035.2	0.00
24	c/D = 3	—	—	8,035.2	0.00
25	c/D = 2	—	—	7,927.2	21.60
26	c/D = 1.5	—	—	7,803.2	43.20
27	c/D = 1.2	—	—	7,091.8	193.99
28	c/D = 1.1	—	—	6,545.6	294.11

Fixed-P curve points — Mx-My contour at Pu = 1,606.2 kN

Cut of the fixed triangulated surface at this combination's own axial force, Design and Nominal paired on the moment direction. The last row repeats the first because the contour is drawn closed.

#	θM (°)	Mx design	My design	M design	Mx nominal	My nominal	M nominal
1	-180.000	-421.75	-0.00	421.75	-528.89	-0.00	528.89
2	-176.857	-398.39	-21.88	398.99	—	—	—
3	-175.570	-391.20	-30.31	392.37	—	—	—
4	-173.864	-380.96	-40.96	383.16	—	—	—
5	-169.825	-356.71	-64.02	362.41	—	—	—
6	-165.357	-331.50	-86.61	342.62	—	—	—
7	-163.470	-323.34	-95.97	337.28	—	—	—
8	-161.653	-315.22	-104.54	332.10	—	—	—
9	-154.918	-286.25	-133.98	316.05	—	—	—
10	-146.340	-249.49	-166.14	299.74	—	—	—
11	-143.904	-239.51	-174.63	296.41	—	—	—
12	-133.846	-197.23	-205.34	284.72	—	—	—
13	-130.639	-184.40	-214.84	283.12	—	—	—
14	-123.128	-152.20	-233.22	278.49	—	—	—
15	-116.296	-123.98	-250.90	279.86	—	—	—
16	-113.504	-111.59	-256.60	279.81	—	—	—
17	-108.651	-90.33	-267.60	282.44	—	—	—
18	-104.895	-73.75	-277.27	286.91	—	—	—
19	-102.946	-65.13	-283.34	290.73	—	—	—
20	-95.475	-28.71	-299.50	300.88	—	—	—
21	-91.677	-9.18	-313.67	313.80	—	—	—
22	-90.879	-4.84	-315.73	315.77	—	—	—
23	-90.000	0.00	-320.08	320.08	0.00	-410.86	410.86
24	-89.121	4.84	-315.73	315.77	—	—	—
25	-88.323	9.18	-313.67	313.80	—	—	—
26	-84.525	28.71	-299.50	300.88	—	—	—
27	-77.054	65.13	-283.34	290.73	—	—	—

#	θ_M (°)	Mx design	My design	M design	Mx nominal	My nominal	M nominal
28	-75.105	73.75	-277.27	286.91	—	—	—
29	-71.349	90.33	-267.60	282.44	—	—	—
30	-66.496	111.59	-256.60	279.81	—	—	—
31	-63.704	123.98	-250.90	279.86	—	—	—
32	-56.872	152.20	-233.22	278.49	—	—	—
33	-49.361	184.40	-214.84	283.12	—	—	—
34	-46.154	197.23	-205.34	284.72	—	—	—
35	-36.096	239.51	-174.63	296.41	—	—	—
36	-33.660	249.49	-166.14	299.74	—	—	—
37	-25.082	286.25	-133.98	316.05	—	—	—
38	-18.347	315.22	-104.54	332.10	—	—	—
39	-16.530	323.34	-95.97	337.28	—	—	—
40	-14.643	331.50	-86.61	342.62	—	—	—
41	-10.175	356.71	-64.02	362.41	—	—	—
42	-6.136	380.96	-40.96	383.16	—	—	—
43	-4.430	391.20	-30.31	392.37	—	—	—
44	-3.143	398.39	-21.88	398.99	—	—	—
45	0.000	421.75	0.00	421.75	528.89	0.00	528.89
46	3.143	398.39	21.88	398.99	—	—	—
47	4.430	391.20	30.31	392.37	—	—	—
48	6.136	380.96	40.96	383.16	—	—	—
49	10.175	356.71	64.02	362.41	—	—	—
50	14.643	331.50	86.61	342.62	—	—	—
51	16.530	323.34	95.97	337.28	—	—	—
52	18.347	315.22	104.54	332.10	—	—	—
53	25.082	286.25	133.98	316.05	—	—	—
54	33.660	249.49	166.14	299.74	—	—	—
55	36.096	239.51	174.63	296.41	—	—	—
56	46.154	197.23	205.34	284.72	—	—	—
57	49.361	184.40	214.84	283.12	—	—	—
58	56.872	152.20	233.22	278.49	—	—	—
59	63.704	123.98	250.90	279.86	—	—	—
60	66.496	111.59	256.60	279.81	—	—	—
61	71.349	90.33	267.60	282.44	—	—	—
62	75.105	73.75	277.27	286.91	—	—	—
63	77.054	65.13	283.34	290.73	—	—	—
64	84.525	28.71	299.50	300.88	—	—	—
65	88.323	9.18	313.67	313.80	—	—	—
66	89.121	4.84	315.73	315.77	—	—	—
67	90.000	0.00	320.08	320.08	0.00	410.86	410.86
68	90.879	-4.84	315.73	315.77	—	—	—
69	91.677	-9.18	313.67	313.80	—	—	—
70	95.475	-28.71	299.50	300.88	—	—	—
71	102.946	-65.13	283.34	290.73	—	—	—
72	104.895	-73.75	277.27	286.91	—	—	—
73	108.651	-90.33	267.60	282.44	—	—	—
74	113.504	-111.59	256.60	279.81	—	—	—
75	116.296	-123.98	250.90	279.86	—	—	—
76	123.128	-152.20	233.22	278.49	—	—	—
77	130.639	-184.40	214.84	283.12	—	—	—
78	133.846	-197.23	205.34	284.72	—	—	—
79	143.904	-239.51	174.63	296.41	—	—	—
80	146.340	-249.49	166.14	299.74	—	—	—
81	154.918	-286.25	133.98	316.05	—	—	—
82	161.653	-315.22	104.54	332.10	—	—	—
83	163.470	-323.34	95.97	337.28	—	—	—

#	θ_M (°)	Mx design	My design	M design	Mx nominal	My nominal	M nominal
84	165.357	-331.50	86.61	342.62	—	—	—
85	169.825	-356.71	64.02	362.41	—	—	—
86	173.864	-380.96	40.96	383.16	—	—	—
87	175.570	-391.20	30.31	392.37	—	—	—
88	176.857	-398.39	21.88	398.99	—	—	—
89	-180.000	-421.75	-0.00	421.75	-528.89	-0.00	528.89
90	-177.382	—	—	—	-520.30	-23.79	520.84
91	-174.692	—	—	—	-513.19	-47.68	515.41
92	-173.026	—	—	—	-506.91	-62.00	510.69
93	-169.209	—	—	—	-492.45	-93.86	501.31
94	-165.349	—	—	—	-471.25	-123.20	487.09
95	-161.907	—	—	—	-451.66	-147.56	475.15
96	-158.405	—	—	—	-429.37	-169.95	461.78
97	-153.022	—	—	—	-397.17	-202.17	445.67
98	-145.161	—	—	—	-348.93	-242.87	425.13
99	-143.822	—	—	—	-340.83	-249.25	422.24
100	-134.949	—	—	—	-287.78	-288.29	407.35
101	-133.164	—	—	—	-277.72	-296.12	405.98
102	-125.015	—	—	—	-228.58	-326.27	398.37
103	-118.728	—	—	—	-191.74	-349.82	398.92
104	-115.339	—	—	—	-170.29	-359.62	397.90
105	-110.740	—	—	—	-141.34	-373.25	399.12
106	-106.557	—	—	—	-114.27	-384.37	401.00
107	-103.700	—	—	—	-95.58	-392.11	403.59
108	-100.675	—	—	—	-74.48	-395.13	402.09
109	-98.719	—	—	—	-61.09	-398.33	402.99
110	-95.457	—	—	—	-38.37	-401.58	403.40
111	-91.213	—	—	—	-8.66	-408.93	409.02
112	-90.622	—	—	—	-4.45	-409.44	409.47
113	-89.378	—	—	—	4.45	-409.44	409.47
114	-88.787	—	—	—	8.66	-408.93	409.02
115	-84.543	—	—	—	38.37	-401.58	403.40
116	-81.281	—	—	—	61.09	-398.33	402.99
117	-79.325	—	—	—	74.48	-395.13	402.09
118	-76.300	—	—	—	95.58	-392.11	403.59
119	-73.443	—	—	—	114.27	-384.37	401.00
120	-69.260	—	—	—	141.34	-373.25	399.12
121	-64.661	—	—	—	170.29	-359.62	397.90
122	-61.272	—	—	—	191.74	-349.82	398.92
123	-54.985	—	—	—	228.58	-326.27	398.37
124	-46.836	—	—	—	277.72	-296.12	405.98
125	-45.051	—	—	—	287.78	-288.29	407.35
126	-36.178	—	—	—	340.83	-249.25	422.24
127	-34.839	—	—	—	348.93	-242.87	425.13
128	-26.978	—	—	—	397.17	-202.17	445.67
129	-21.595	—	—	—	429.37	-169.95	461.78
130	-18.093	—	—	—	451.66	-147.56	475.15
131	-14.651	—	—	—	471.25	-123.20	487.09
132	-10.791	—	—	—	492.45	-93.86	501.31
133	-6.974	—	—	—	506.91	-62.00	510.69
134	-5.308	—	—	—	513.19	-47.68	515.41
135	-2.618	—	—	—	520.30	-23.79	520.84
136	2.618	—	—	—	520.30	23.79	520.84
137	5.308	—	—	—	513.19	47.68	515.41
138	6.974	—	—	—	506.91	62.00	510.69
139	10.791	—	—	—	492.45	93.86	501.31

#	θ_M (°)	Mx design	My design	M design	Mx nominal	My nominal	M nominal
140	14.651	—	—	—	471.25	123.20	487.09
141	18.093	—	—	—	451.66	147.56	475.15
142	21.595	—	—	—	429.37	169.95	461.78
143	26.978	—	—	—	397.17	202.17	445.67
144	34.839	—	—	—	348.93	242.87	425.13
145	36.178	—	—	—	340.83	249.25	422.24
146	45.051	—	—	—	287.78	288.29	407.35
147	46.836	—	—	—	277.72	296.12	405.98
148	54.985	—	—	—	228.58	326.27	398.37
149	61.272	—	—	—	191.74	349.82	398.92
150	64.661	—	—	—	170.29	359.62	397.90
151	69.260	—	—	—	141.34	373.25	399.12
152	73.443	—	—	—	114.27	384.37	401.00
153	76.300	—	—	—	95.58	392.11	403.59
154	79.325	—	—	—	74.48	395.13	402.09
155	81.281	—	—	—	61.09	398.33	402.99
156	84.543	—	—	—	38.37	401.58	403.40
157	88.787	—	—	—	8.66	408.93	409.02
158	89.378	—	—	—	4.45	409.44	409.47
159	90.622	—	—	—	-4.45	409.44	409.47
160	91.213	—	—	—	-8.66	408.93	409.02
161	95.457	—	—	—	-38.37	401.58	403.40
162	98.719	—	—	—	-61.09	398.33	402.99
163	100.675	—	—	—	-74.48	395.13	402.09
164	103.700	—	—	—	-95.58	392.11	403.59
165	106.557	—	—	—	-114.27	384.37	401.00
166	110.740	—	—	—	-141.34	373.25	399.12
167	115.339	—	—	—	-170.29	359.62	397.90
168	118.728	—	—	—	-191.74	349.82	398.92
169	125.015	—	—	—	-228.58	326.27	398.37
170	133.164	—	—	—	-277.72	296.12	405.98
171	134.949	—	—	—	-287.78	288.29	407.35
172	143.822	—	—	—	-340.83	249.25	422.24
173	145.161	—	—	—	-348.93	242.87	425.13
174	153.022	—	—	—	-397.17	202.17	445.67
175	158.405	—	—	—	-429.37	169.95	461.78
176	161.907	—	—	—	-451.66	147.56	475.15
177	165.349	—	—	—	-471.25	123.20	487.09
178	169.209	—	—	—	-492.45	93.86	501.31
179	173.026	—	—	—	-506.91	62.00	510.69
180	174.692	—	—	—	-513.19	47.68	515.41
181	177.382	—	—	—	-520.30	23.79	520.84